KopiKaki

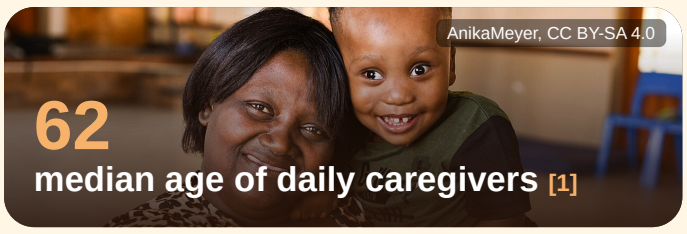
A voice-first AI friend that helps older caregivers with failing eyesight finish online support forms.

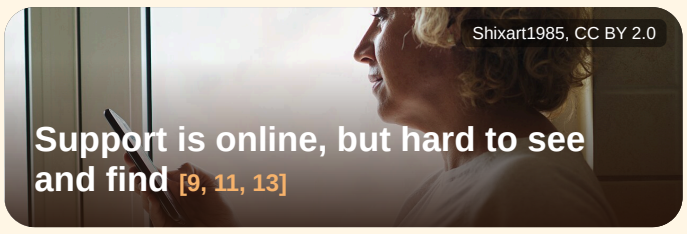
WHO Older caregivers with declining vision

TASK Applying for a caregiver support grant online

AI'S JOB Read, explain, fill in. Never submit.

1 THE PROBLEM





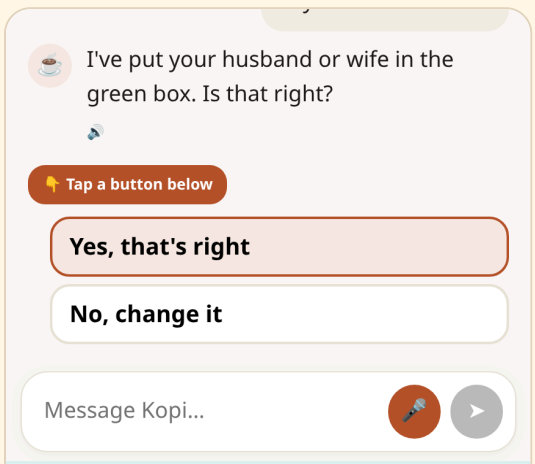


Older caregivers with declining vision give up on online support forms, so the help meant for them never arrives.

WHAT EXISTS TODAY
General digital-skills programmes for seniors, plus agency hotlines [14]

THE GAP
No caregiving-specific help at the moment someone is stuck on the form [14]

2 THE SOLUTION: KOPI READS, EXPLAINS, FILLS IN. YOU DECIDE.



Working prototype. Listens first (ELIZA-style), then guides the form **one question at a time**, lights up the matching box and reads every entry back.

Offers help [3, 6] → **Reads aloud** [9] → **Explains simply** [11, 13] → **Fills in, reads back** [11] → **You press Submit** [8]

WHAT IT IS & WHO IT IS FOR
A chat companion beside an agency form, for older caregivers whose eyesight makes dense forms hard to read. Offered by community partners, never forced [3, 6, 9, 10]. Speaks English, Mandarin, Malay and Tamil.

WHAT THE AI DOES
- ✓ Reads and explains each section in plain words
- ✓ Listens to typed or spoken answers
- ✓ Fills the non-sensitive boxes, reads each back
- ✓ Spots scam wording

WHAT THE AI NEVER DOES
- ✕ Types NRIC, income or bank details
- ✕ Presses Submit for you
- ✕ Predicts how much you will get
- ✕ Guesses when unsure: it hands over

3 INTERDISCIPLINARY FRAMING: THREE LENSES, ONE DESIGN

STEM
Speech and screen-reading AI for low vision [9, 11]

BUSINESS & GOVERNANCE
Works inside agency rules; saves staff time [14-18]

SOCIAL SCIENCES
Dignity: caregiving is seen as personal duty, help is rarely asked for [3, 5, 6]

WHERE THE LENSES MEET: EACH DESIGN RULE COMES FROM TWO OF THEM

SOCIETY × GOVERNANCE
Dignity + consent: the user types private details and presses Submit. Kopi never acts alone.

STEM × SOCIETY
Low vision + not wanting to ask: voice-first, one lit-up box at a time, offered but never forced.

GOVERNANCE × STEM
Agency data rules: Kopi sees only the form, keeps nothing, and Singpass stays with the user [14].

4 HUMAN-IN-THE-LOOP PROTOCOL proposed

WHEN	KOPI STOPS AND	THE HUMAN
Not understood twice, or asks for a person	Says "I'm not sure" and hands over [8]	**Hotline officer takes over**
NRIC, income, bank, payment	Masks it, tells the user to type it in the form	**The user, alone**
25 seconds of silence	Pauses; nothing is sent	**The user taps "Carry on"**
Scam wording (fee, OTP)	Halts and warns; flags to agency	**Agency reviews the flag**
Self-harm or abuse words	Stops chatting, gives hotline numbers	**Trained responder**

5 STAKEHOLDER CONCERNS

Feasibility	No agency system changes: Kopi sits beside the public form. Prototype built; the pilot tests it.
Cost	Not yet costed, and we say so. The pilot records staff time and cost per completed form.
Ethics & privacy	Opt-in, on-device, deleted after use; the user always has the final say. We test across languages and ages for bias.
Implementation	Free setup by hospitals, CCs and volunteers [9, 10]. If Kopi is wrong, it says "not sure".

6 WHY SOMEONE SHOULD ACT

Caregivers	Finish it themselves, with dignity. **Ask:** try it.
Agencies	More schemes reach the people they are for [13, 14]. **Ask:** let Kopi sit beside your form.
CCs & hospitals	A tool for outreach they already do [9, 10]. **Ask:** host the set-up.

7 PILOT

1 community club · ~30 caregivers · 3 months. Measure: forms abandoned · hotline calls · confidence · loneliness · cost per form [5, 8, 11, 13]

[1] Tan et al. (2025), SMU. **[3]** Tan et al. (2026), The Straits Times. **[5]** Yong et al. (2025), Gerontology 71(11). **[6]** Lim-Soh et al. (2024), J Gerontol B 80(1). **[8]** Liu et al. (2026), Front Public Health 14. **[9]** Visaria et al. (2023), Aging Health Res 3(1). **[10]** Gao et al. (2025), Health Soc Care Community. **[11]** Hassan (2020), JMIR Aging 3(2). **[13]** Seah et al. (2025), J Aging Soc Policy. **[14]** Ministry of Health (2023), Action Plan for Successful Ageing. **[15]** Zhan & Huang (2023), China Popul Dev Stud 7. **[16]** Moriyama (2026), J Econ Ageing 34. **[17]** Azman et al. (2024), Aging Health Res 4(2). **[18]** Keita Fakeye et al. (2023), Value Health 26(5).

Photos (illustrative, not Singapore caregivers except the coffee shop): Wikimedia Commons: “A woman and her grandson…” by AnikaMeyer (CC BY-SA 4.0); “Elderly woman standing next to a window…” and “Elderly woman with a laptop…” by Shixart1985 (CC BY 2.0); “CoffeeShopSG” by mailer_diablo (CC BY-SA 3.0).